

```
!pip install langchain langchain_google_genai langchain_community duckduckgo-search
!pip install -U ddgs
```

```
Requirement already satisfied: langchain in /usr/local/lib/python3.12/dist-packages (0.3.27)
Requirement already satisfied: langchain_google_genai in /usr/local/lib/python3.12/dist-packages (2.1.12)
Requirement already satisfied: langchain_community in /usr/local/lib/python3.12/dist-packages (0.3.30)
Requirement already satisfied: duckduckgo-search in /usr/local/lib/python3.12/dist-packages (8.1.1)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.72 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.3.72)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.3.9)
Requirement already satisfied: langsmith>=0.1.17 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.4.31)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.11.9)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.0.43)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.32.5)
Requirement already satisfied: PyYAML<=5.3 in /usr/local/lib/python3.12/dist-packages (from langchain) (6.0.3)
Requirement already satisfied: google-ai-generative-language<1,>=0.7 in /usr/local/lib/python3.12/dist-packages (from langchain-google-genai) (0.7.0)
Requirement already satisfied: filetype<2,>=1.2 in /usr/local/lib/python3.12/dist-packages (from langchain-google-genai) (1.2.0)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (3.10.10)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (9.0.0)
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (0.6.7)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.10.1)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (0.4.0)
Requirement already satisfied: numpy<=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.0.2)
Requirement already satisfied: click>=8.1.8 in /usr/local/lib/python3.12/dist-packages (from duckduckgo-search) (8.3.0)
Requirement already satisfied: primp>=0.15.0 in /usr/local/lib/python3.12/dist-packages (from duckduckgo-search) (0.15.0)
Requirement already satisfied: lxml>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from duckduckgo-search) (6.0.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (6.3.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (0.2.0)
Requirement already satisfied: yarll<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.17.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json) (3.22.0)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json) (0.9.0)
Requirement already satisfied: google-api-core!=2.0.*,!=2.1.*,!=2.10.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,!=2.8.* in /usr/local/lib/python3.12/dist-packages (from google-ai-generative-language) (2.24.0)
Requirement already satisfied: google-auth!=2.24.0,!>2.25.0,<3.0.0,>=2.14.1 in /usr/local/lib/python3.12/dist-packages (from google-ai-generative-language) (2.36.0)
Requirement already satisfied: proto-plus<2.0.0,>=1.22.3 in /usr/local/lib/python3.12/dist-packages (from google-ai-generative-language) (1.25.0)
Requirement already satisfied: protobuf!=4.21.0,!>4.21.1,!>4.21.2,!>4.21.3,!>4.21.4,!>4.21.5,<7.0.0,>=3.20.2 in /usr/local/lib/python3.12/dist-packages (from google-ai-generative-language) (4.21.5)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core) (1.33.0)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core) (4.12.0)
Requirement already satisfied: packaging<26.0.0,>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core) (25.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith) (0.27.0)
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith) (3.10.14)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith) (1.0.0)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith) (0.23.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic) (0.7.0)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2025.10.1)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from SQLAlchemy) (3.1.1)
Requirement already satisfied: googleapis-common-protos<2.0.0,>=1.56.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core) (1.66.0)
Requirement already satisfied: grpcio<2.0.0,>=1.33.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core[grpc]) (1.66.2)
Requirement already satisfied: grpcio-status<2.0.0,>=1.33.2 in /usr/local/lib/python3.12/dist-packages (from google-api-core[grpc]) (1.66.2)
Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from google-auth) (5.5.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth) (0.6.1)
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth) (4.9.1)
```

```
import os
from langchain_google_genai import ChatGoogleGenerativeAI
from langchain_community.tools import DuckDuckGoSearchResults
from langchain.tools import Tool
from langchain.agents import initialize_agent, AgentType

# =====
# 🗝 Step 1: Authenticate Gemini using API key
# =====
# Get your Gemini API key from https://aistudio.google.com/app/apikey
os.environ["GOOGLE_API_KEY"] = "AIzaSyAcj0or3UwW4YK3EEJ3cx4BwqmtFdj1vwc"
```

```
# 🧠 Step 2: Initialize Gemini 2.5 Flash Model
# =====
llm = ChatGoogleGenerativeAI(model="gemini-2.5-flash", temperature=0.3)

# =====
# 🌐 Step 3: Define Search Tool (DuckDuckGo)
# =====
search_tool = DuckDuckGoSearchResults(name="Web Search")
```

```
def summarize_text(query: str) -> str:
    """Summarize text or search results using Gemini."""
    prompt = f"Summarize the following text into 5 short, clear bullet points:\n\n{query}"
    response = llm.invoke(prompt)
    return response.content if hasattr(response, "content") else str(response)

summarize_tool = Tool(
    name="Summarizer",
    func=summarize_text,
    description="Summarizes text or content using Gemini model."
)
```

```
# 🌟 Step 5: Create an Agent with Both Tools
# =====
tools = [search_tool, summarize_tool]
agent = initialize_agent(
    tools,
    llm,
    agent_type=AgentType.ZERO_SHOT_REACT_DESCRIPTION,
    verbose=True
)
```

```
/tmp/ipython-input-293671903.py:4: LangChainDeprecationWarning: LangChain agents will continue to be supported, but it is recom
agent = initialize_agent(
```

```
query = "Summarize the latest advancements in Generative AI"
response = agent.run(query)
```

```
print("\n✅ FINAL SUMMARY:\n", response)
```

```
/tmp/ipython-input-2771596875.py:2: LangChainDeprecationWarning: The method `Chain.run` was deprecated in langchain 0.1.0 and v
response = agent.run(query)
```

```
> Entering new AgentExecutor chain...
```

```
Action: Web Search
```

```
Action Input: Latest advancements in Generative AI
```

```
Observation: snippet: Théâtre D'opéra Spatial, an image made using generative AI . Generative artificial intelligence is a suby
Thought:Action: Web Search
```

```
Action Input: recent breakthroughs in Generative AI
```

```
Observation: snippet: ... Generative AI use cases, which we ... Generative AI , and cognitive automation in general, is shorter
Thought:Action: Web Search
```

```
Action Input: "recent advancements in generative AI" 2023 2024
```

```
Observation: snippet: Théâtre D'opéra Spatial, an image made using generative AI . Generative artificial intelligence is a suby
Thought:Action: Web Search
```

```
Action Input: "summary of recent generative AI advancements" 2023 2024
```

```
Observation: snippet: This summary generator tool takes long texts and makes them shorter and clearer. Our AI summarizer reads
Thought:Action: Web Search
```

```
Action Input: "key advancements in generative AI" 2023 2024
```

```
Observation: snippet: 1 day ago - In December 2023, Google unveiled Gemini, a multimodal AI model available in four versions: l
Thought:Thought:I have gathered information from several web searches regarding the Latest advancements in Generative AI in 2023 and 2024.
Final Answer:The Latest advancements in Generative AI during 2023 and 2024 have been characterized by rapid development and widespread integration.
```

```
* **Multimodal AI Models:** The introduction of advanced multimodal models like Google's Gemini (Ultra, Pro, Flash, Nano) in
* **Integration and Application Expansion:** Generative AI is increasingly being integrated into everyday tools and services,
* **Economic Impact:** Reports from firms like McKinsey & Company and Goldman Sachs Research project a massive economic impac
* **Focus on Efficiency and Ethics:** Future trends emphasize model efficiency, the development of instant applications, and
* **Rapid Pace of Innovation:** The field continues to evolve at an unprecedented pace, with ongoing research and development
```

```
> Finished chain.
```

```
✅ FINAL SUMMARY:
```

```
The latest advancements in Generative AI during 2023 and 2024 have been characterized by rapid development and widespread inte
```

```
* **Multimodal AI Models:** The introduction of advanced multimodal models like Google's Gemini (Ultra, Pro, Flash, Nano) in
* **Integration and Application Expansion:** Generative AI is increasingly being integrated into everyday tools and services,
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Start coding or [generate](#) with AI.

